

Table 2: Descriptive statistics on two-way fixed effects papers

	# Papers
<i>Panel A. Estimation method</i>	
Fixed-effects OLS regression	13
First-difference OLS regression	6
Fixed-effects or first-difference OLS regression, with several treatment variables	6
Fixed-effects or first-difference 2LS regression	3
Other regression	5
<i>Panel B. Research design</i>	
Sharp design	26
Fuzzy design	7
<i>Panel C. Are there stable groups?</i>	
Yes	12
Presumably yes	14
Presumably no	5
No	2

Notes. This table reports the estimation method and the research design used in the 33 papers using two-way fixed effects regressions published in the AER from 2010 to 2012, and whether those papers have stable groups.

6.2 Application to Gentzkow et al. (2011)

Gentzkow et al. (2011) study the effect of newspapers on voters' turnout in US presidential elections between 1868 and 1928. They regress the first-difference of the turnout rate in county g between election years $t - 1$ and t on state-year fixed effects and on the first difference of the number of newspapers available in that county. This corresponds to Regression 2, with state-year fixed effects as controls. As reproduced in Table 3 below, Gentzkow et al. (2011) find that $\hat{\beta}_{fd} = 0.0026$ (s.e. = 9×10^{-4}). According to this regression, one more newspaper increased voters' turnout by 0.26 percentage points. On the other hand, $\hat{\beta}_{fe} = -0.0011$ (s.e. = 0.0011). $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ are significantly different (t-stat=2.86).

We use the `twowayfeweights` Stata package, downloadable with its help file from the SSC repository, to estimate the weights attached to $\hat{\beta}_{fe}$. 6,212 are strictly positive, 4,161 are strictly negative. The negative weights sum to -0.53. $\hat{\sigma}_{fe} = 3 \times 10^{-4}$, meaning that β_{fe} and the ATT may be of opposite signs if the standard deviation of the ATEs across all the treated (g, t) cells

is equal to 0.0003.¹⁸ $\hat{\underline{\sigma}}_{fe} = 7 \times 10^{-4}$, meaning that β_{fe} may be of a different sign than the ATEs of all the treated (g, t) cells if the standard deviation of those ATEs is equal to 0.0007. We also estimate the weights attached to $\hat{\beta}_{fd}$. 5,472 are strictly positive, and 4,605 are strictly negative. The negative weights sum to -1.43. $\hat{\underline{\sigma}}_{fd} = 4 \times 10^{-4}$, and $\hat{\underline{\sigma}}_{fd} = 6 \times 10^{-4}$.

Therefore, β_{fe} and β_{fd} can only receive a causal interpretation if the weights attached to them are uncorrelated with the intensity of the treatment effect in each county $\times$ election-year cell (Assumptions 7 and 8, respectively). This is not warranted. First, as $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ significantly differ, Assumptions 7 and 8 cannot jointly hold. Moreover, the weights attached to $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ are correlated with variables that are likely to be themselves associated with the intensity of the treatment effect in each cell. For instance, the correlation between the weights attached to $\hat{\beta}_{fd}$ and t , the year variable, is equal to -0.06 (t-stat=-3.28). The effect of newspapers may be different in the last than in the first years of the panel. For instance, new means of communication, like the radio, appear in the end of the period under consideration, and may diminish the effect of newspapers.¹⁹ This would lead to a violation of Assumption 8.

The stable groups assumption holds: between each pair of consecutive elections, there are counties where the number of newspapers does not change. We use the `fuzzydid` Stata package, downloadable with its help file from the SSC repository, to estimate a modified version of our DID_M estimator, that accounts for the fact that the number of newspapers is not binary (see section ?? of our Web Appendix, where we define this modified estimator). We include state-year fixed effects as controls in our estimation. We find that DID_M = 0.0043, with a standard error of 0.0015. DID_M is 66% larger than $\hat{\beta}_{fd}$, and the two estimators are significantly different at the 10% level (t-stat=1.69). DID_M is also of a different sign than $\hat{\beta}_{fe}$.

Our DID_M estimator only relies on a common trends assumption. To assess its plausibility, we compute DID_M^{pl}, the placebo estimator introduced in Section 4.²⁰ As shown in Table 3 below, our placebo estimator is small and not significantly different from 0, meaning that counties where the number of newspapers increased or decreased between $t - 1$ and t did not experience significantly different trends in turnout from $t - 2$ to $t - 1$ than counties where that number was stable. Our placebo estimator is estimated on a subset of the data: for each pair of consecutive time periods $t - 1$ and t , we only keep counties where the number of newspapers did not change between $t - 2$ and $t - 1$. Still, almost 80% of the county $\times$ election-year observations are used in the computation of the placebo estimator. Moreover, when reestimated on this subsample, the DID_M estimator is very close to the DID_M estimator in the full sample.

¹⁸The number of newspapers is not binary, so strictly speaking, in this application the parameter of interest is the average causal response parameter introduced in Section ?? of our Web Appendix, rather than the ATT.

¹⁹In fact, Gentzkow et al. (2011) analyze the 1868 to 1928 period separately from later periods, because the growth of the radio may have changed newspapers' effects.

²⁰Again, we need to slightly modify DID_M^{pl} to account for the fact that the number of newspapers is not binary.

Table 3: Estimates of the effect of one additional newspaper on turnout

	Estimate	Standard error	N
$\hat{\beta}_{fd}$	0.0026	0.0009	15,627
$\hat{\beta}_{fe}$	-0.0011	0.0011	16,872
DID _M	0.0043	0.0015	16,872
DID _M ^{pl}	-0.0009	0.0016	13,221
DID _M , on placebo subsample	0.0045	0.0019	13,221

Notes. This table reports estimates of the effect of one additional newspaper on turnout, as well as a placebo estimate of the common trends assumption underlying DID_M. Estimators are computed using the data of Gentzkow et al. (2011), with state-year fixed effects as controls. Standard errors are clustered by county. To compute the DID_M estimators, the number of newspapers is grouped into 4 categories: 0, 1, 2, and more than 3.

6.3 The effect of union membership on wages

A number of articles have estimated the effect of union membership on wages using panel data and controlling for workers' fixed effects. For instance, Jakubson (1991) has found a 8.3% union membership premium using that strategy, in a sample of American males from the PSID followed from 1976 to 1980. Vella and Verbeek (1998) estimate a similar regression and find similar results, in a sample of young American males from the NLSY followed from 1980 to 1987.²¹

We use the data in Vella and Verbeek (1998) to compute various estimators of the union wage premium. As union status is often measured with error (see, e.g. Freeman, 1984; Card, 1996), we discard changes in union status happening twice in three consecutive years. Specifically, for individuals with $D_{i,t-1} = 0$, $D_{i,t} = 1$, and $D_{i,t+1} = 0$, we replace $D_{i,t}$ by 0. Similarly, for individuals with $D_{i,t-1} = 1$, $D_{i,t} = 0$, and $D_{i,t+1} = 1$, we replace $D_{i,t}$ by 1. Doing so, we discard half of the union status changes in the initial data.²²

We start by estimating a two-way fixed effects regression of wages on union membership with worker and year fixed effects. Table 4 below shows that $\hat{\beta}_{fe} = 0.107$ (s.e. = 0.030), a result close to that of the worker fixed effects regressions in Jakubson (1991) and Vella and Verbeek (1998).

Then, we estimate the weights attached to $\hat{\beta}_{fe}$. 820 are strictly positive, 196 are strictly negative, but the negative weights only sum to -0.01. Still, $\hat{\sigma}_{fe} = 0.097$, meaning that β_{fe} and the ATT may be of opposite signs if the standard deviation of the treatment effect across the unionized worker $\times$ year observations is equal to 0.097, a substantial but still possible amount of heterogeneity.

²¹The fixed effects regression is not the main specification in Vella and Verbeek (1998). The authors favor instead a dynamic selection model.

²² Keeping the original data does not change much the results presented below, except that the placebo estimator DID_M^{pl,2} becomes significant.

The weights are negatively correlated with workers' years of schooling (correlation = -0.12 , t-stat = -1.88). The union premium may be lower for more educated workers (see Freeman and Medoff, 1984), as they may be less substitutable than less educated ones. Then, $\hat{\beta}_{fe}$ may overestimate δ^{TR} , the average union premium across all unionized worker $\times$ year observations. We also find that $\hat{\beta}_{fd} = 0.060$ (s.e. = 0.032) and that $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ significantly differ (t-stat = 1.91),²³ thus casting further doubt on Assumptions 7 and 8.

The stable groups assumption holds: between each pair of consecutive years, there are workers whose union membership status does not change. We therefore compute our DID_M estimator. Table 4 shows that it is equal to 0.041 (s.e. = 0.034). DID_M is significantly different from $\hat{\beta}_{fe}$ (t-stat = 2.60) and $\hat{\beta}_{fd}$ (t-stat = 2.36).²⁴ As discussed in Section 4, we can also estimate separately the union premium for workers joining and leaving a union, something that was previously done by Freeman (1984). The joiners' effect estimate is equal to 0.059 (s.e. = 0.053), the leavers' effect is equal to 0.021 (s.e. = 0.044), and the two estimates do not significantly differ (t-stat = 0.55).

DID_M relies on a common trends assumption. To assess its plausibility, we compute DID_M^{pl}, the placebo estimator introduced in Section 4. DID_M^{pl} compares the wage growth of workers changing and not changing their union status one period before that change. We also compute DID_M^{pl,2} and DID_M^{pl,3}, two other placebo estimators performing the same comparison two and three periods before the change. As shown in Table 4 below, DID_M^{pl} is large, positive, and significant (t-stat = 2.49). On the other hand DID_M^{pl,2} and DID_M^{pl,3} are smaller and insignificant. Workers that become unionized start experiencing a differential positive pre-trend one year before becoming unionized. This differential pre-trend mostly comes from union joiners: for them, the placebo estimator is equal to 0.119 (s.e. = 0.051), while for union leavers the placebo is smaller (0.061) and insignificant (s.e. = 0.057). Therefore, the placebos suggest that even the already small and insignificant DID_M estimator may overestimate the union premium, due to a positive pre-trend. In fact, the estimate of leavers' effect, for which there is no evidence of a pre-trend, is very close to 0. Overall, our results indicate that there may not be a significant union wage premium.

²³The standard error of $\hat{\beta}_{fe} - \hat{\beta}_{fd}$ is computed with a worker-level clustered bootstrap.

²⁴The standard errors of $\hat{\beta}_{fe} - \text{DID}_M$ and $\hat{\beta}_{fd} - \text{DID}_M$ are computed with a worker-level clustered bootstrap.